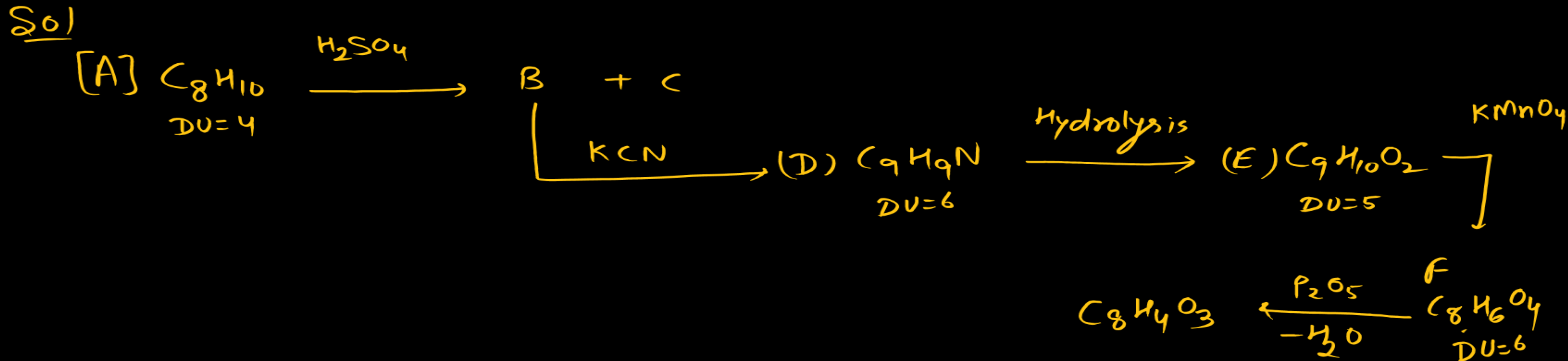


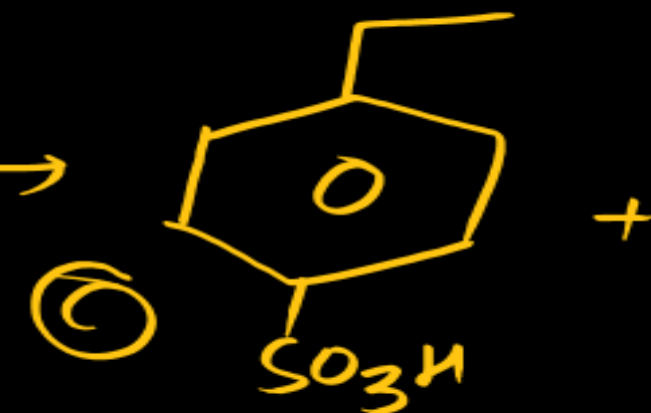
Q. An organic compound $A(C_8H_{10})$ on treatment with fuming sulphuric acid yields two isomeric products B and C . B on fusion with KCN gives $D(C_9H_9N)$ which on hydrolysis yields $E(C_9H_{10}O_2)$. E on treatment with hot alkaline $KMnO_4$ yields $F(C_8H_6O_4)$. F on heating with P_2O_5 undergo intramolecular dehydration to yield $G(C_8H_4O_3)$. Identify A to G .



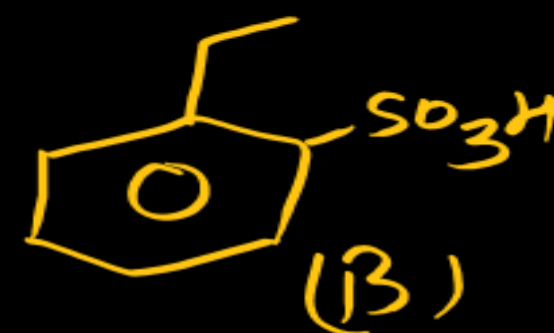
[A]



conc.
 H_2SO_4



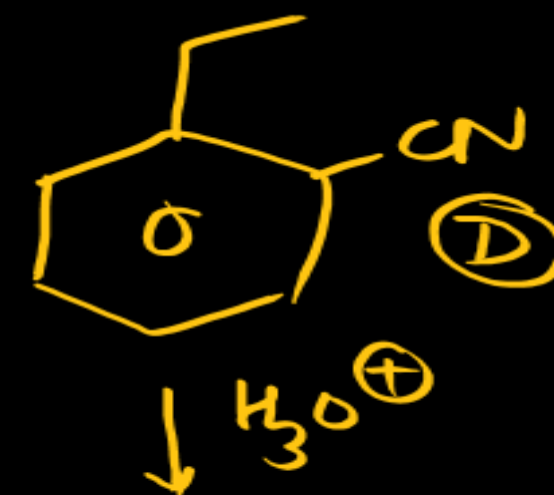
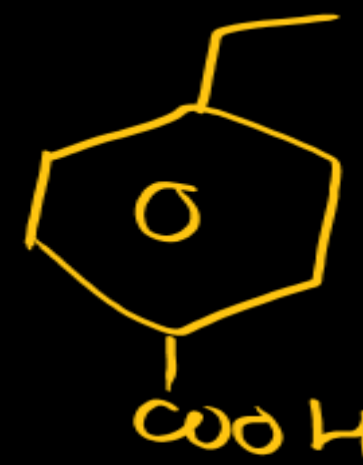
+



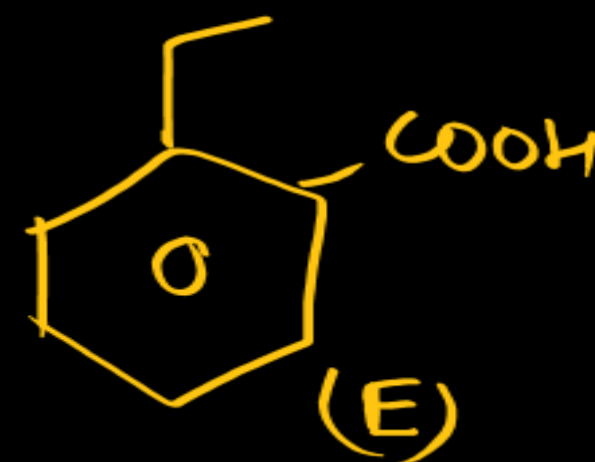
KCN



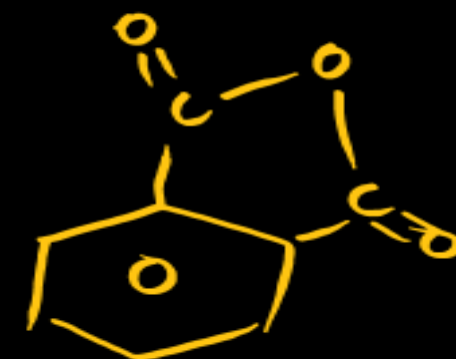
H_3O^+



H_3O^+

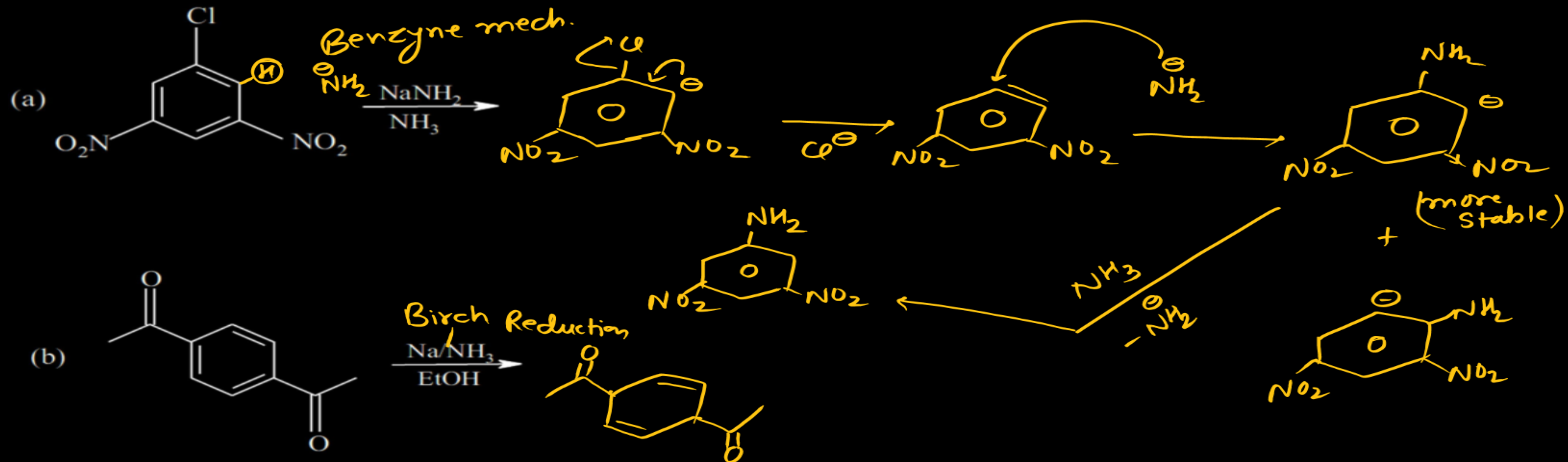


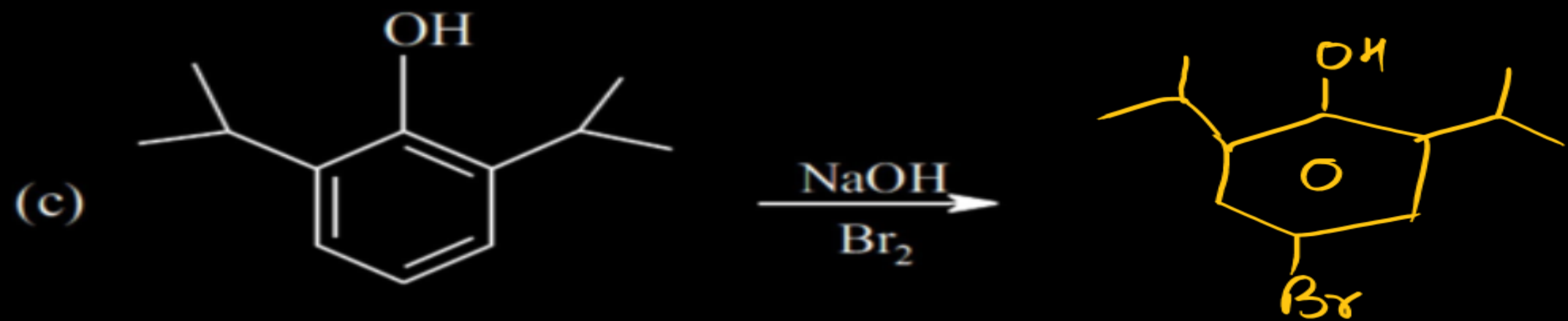
$KMnO_4$

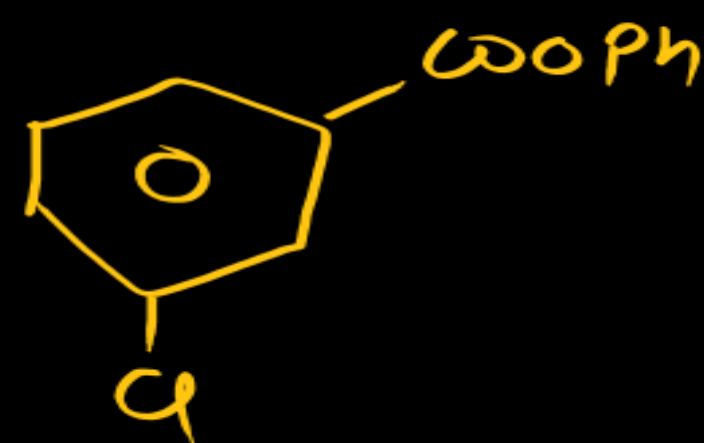
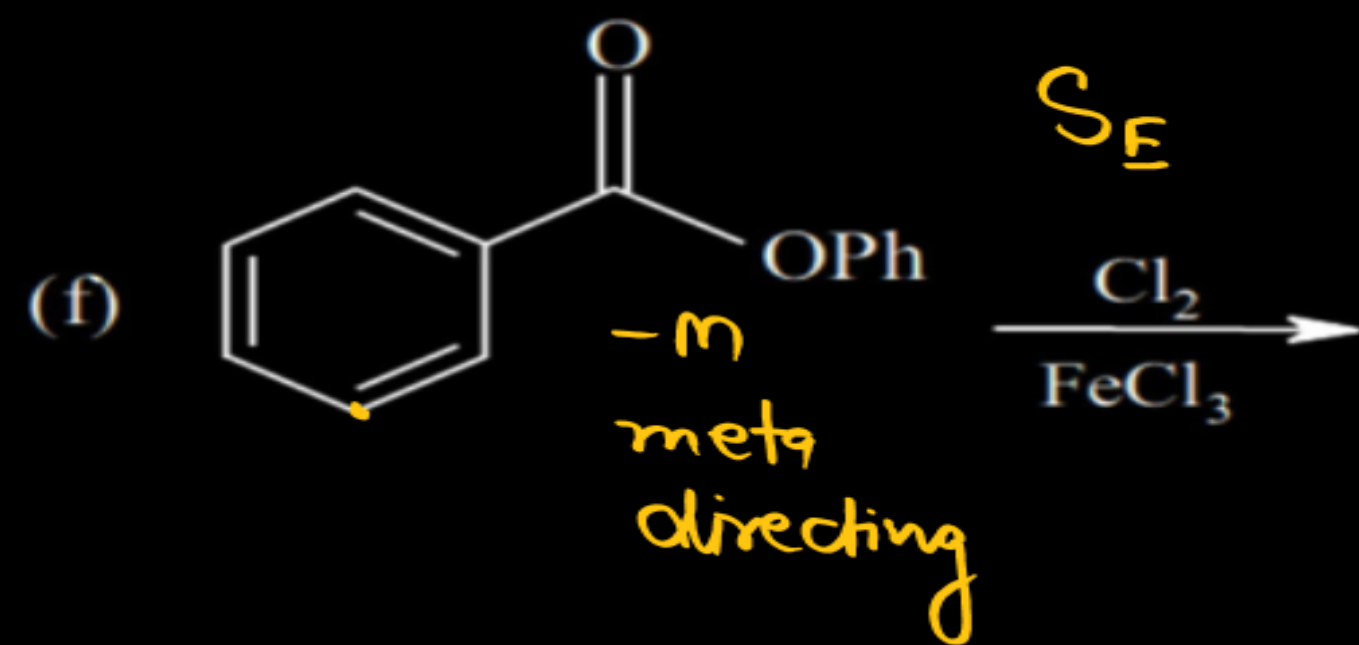
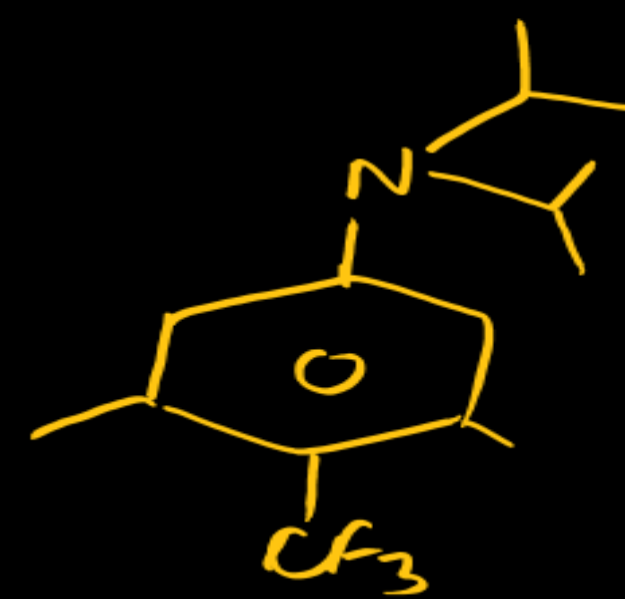
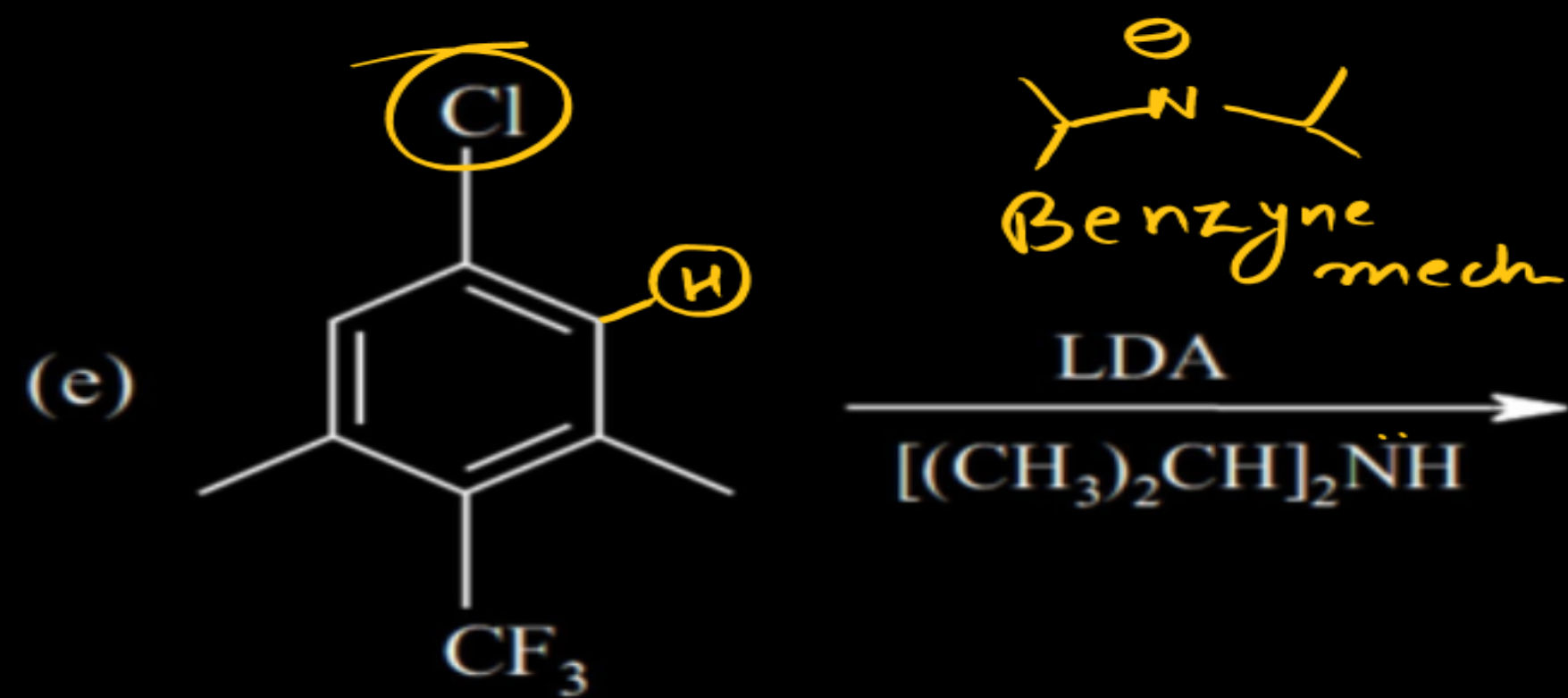


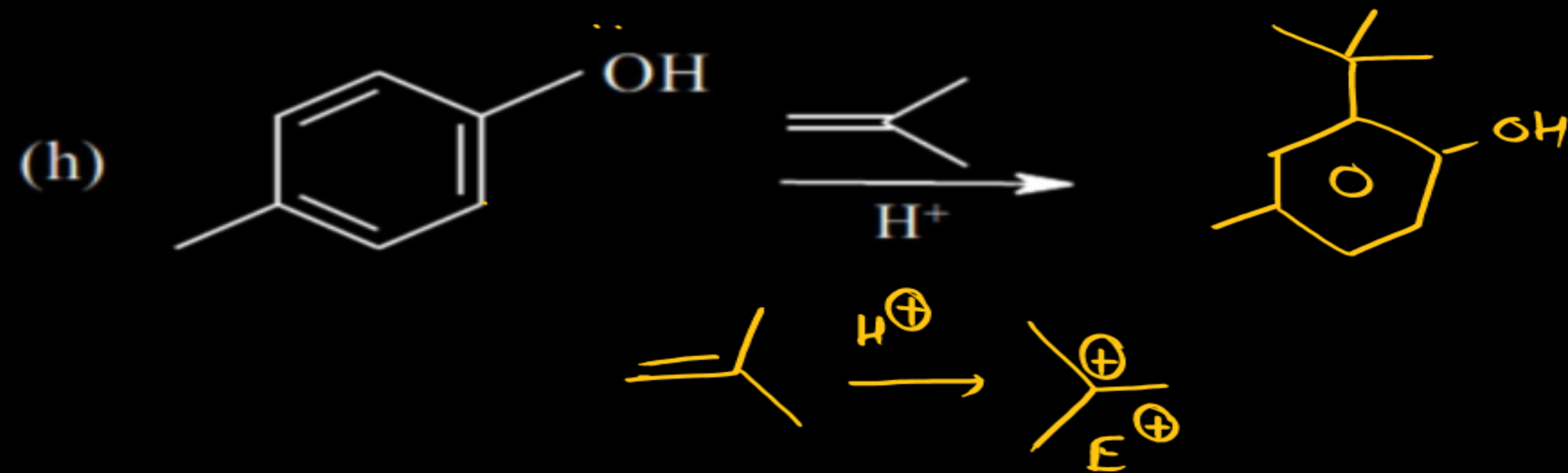
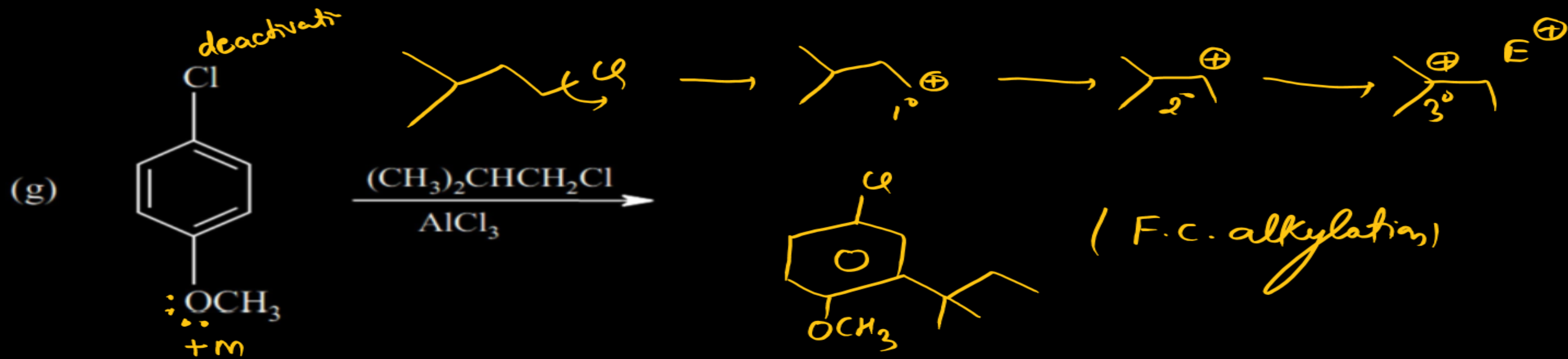
P_2O_5
 $-H_2O$

Q. Predict product of the following reactions:

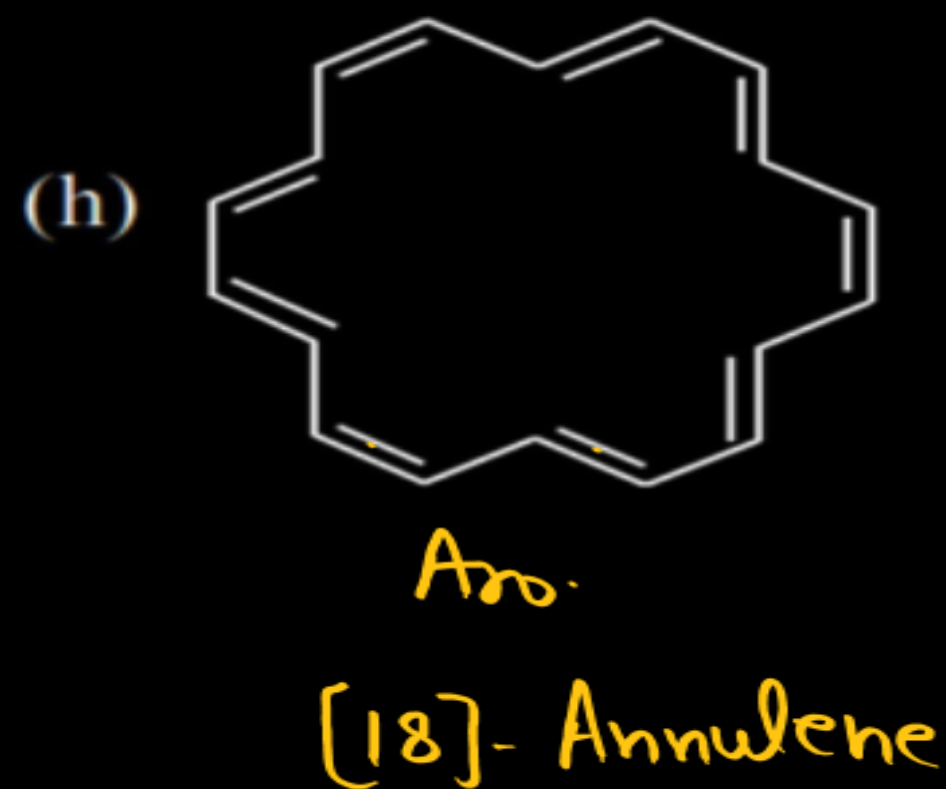
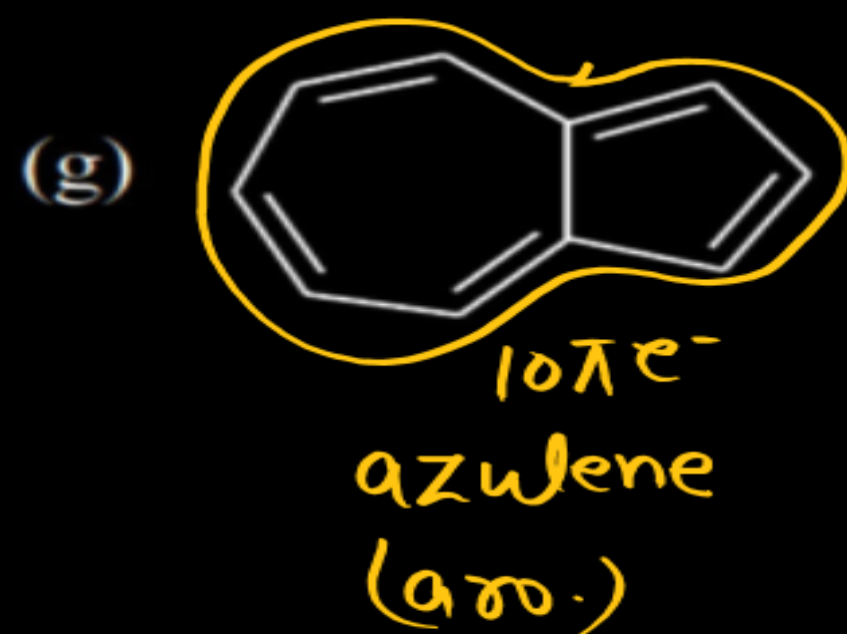
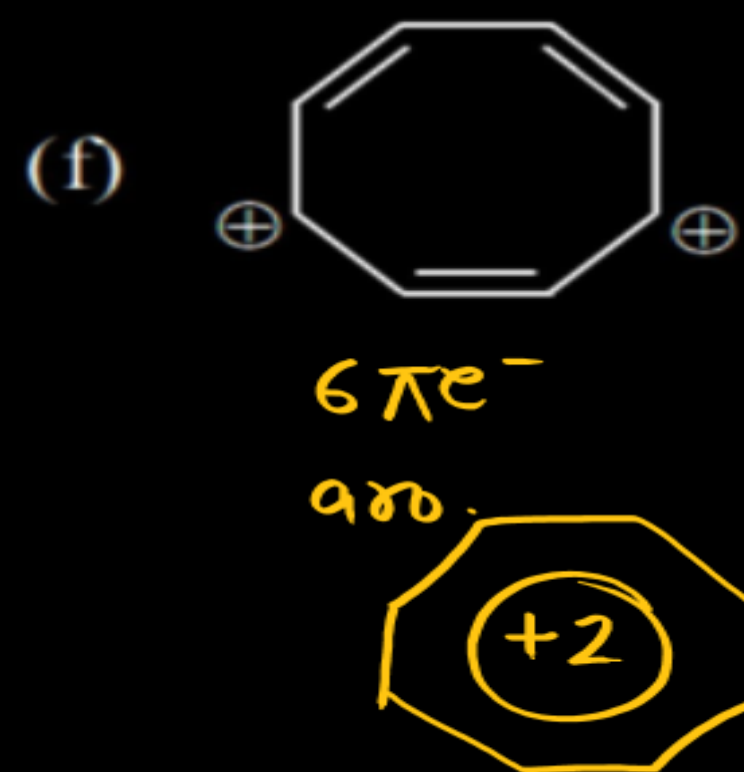
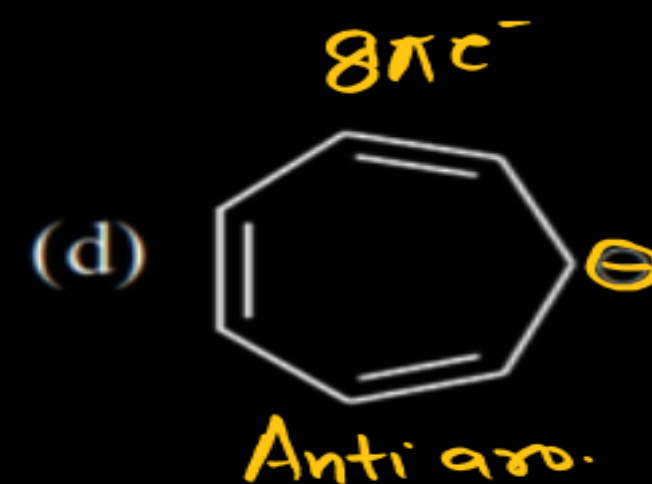
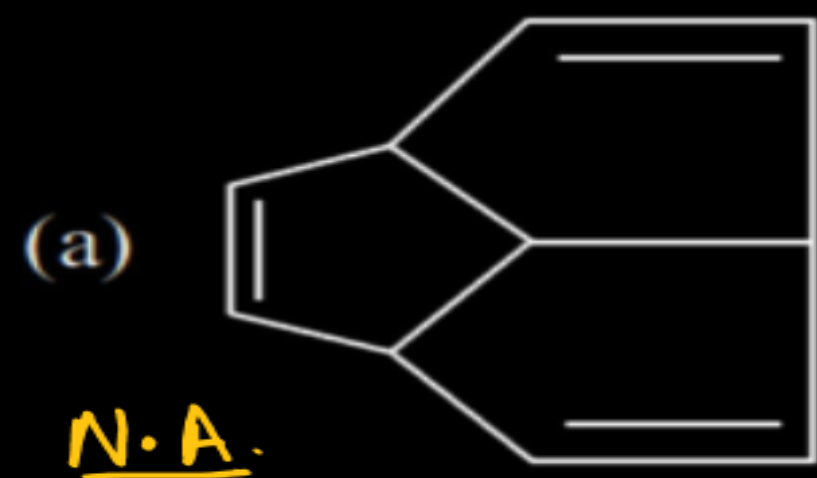








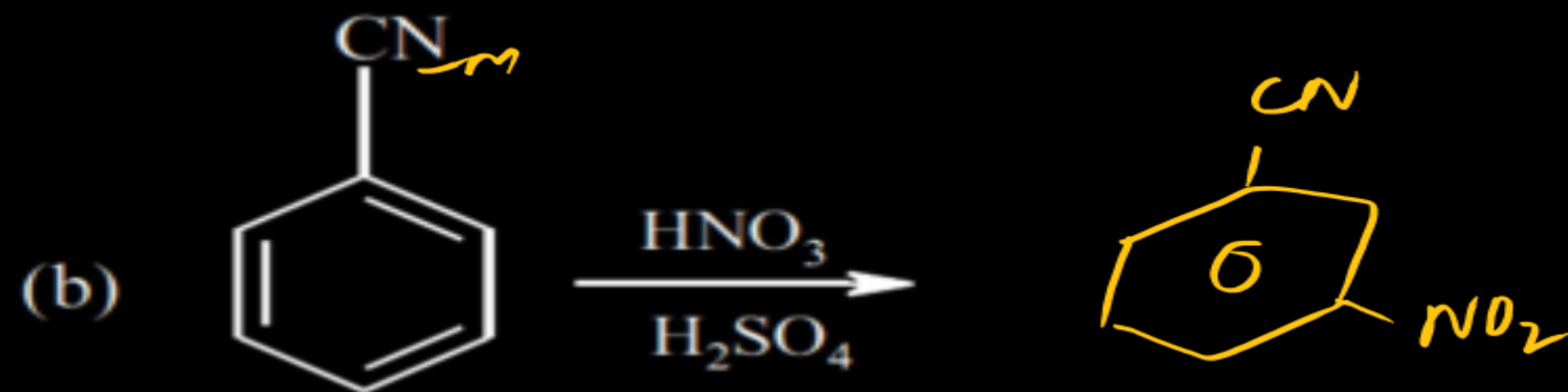
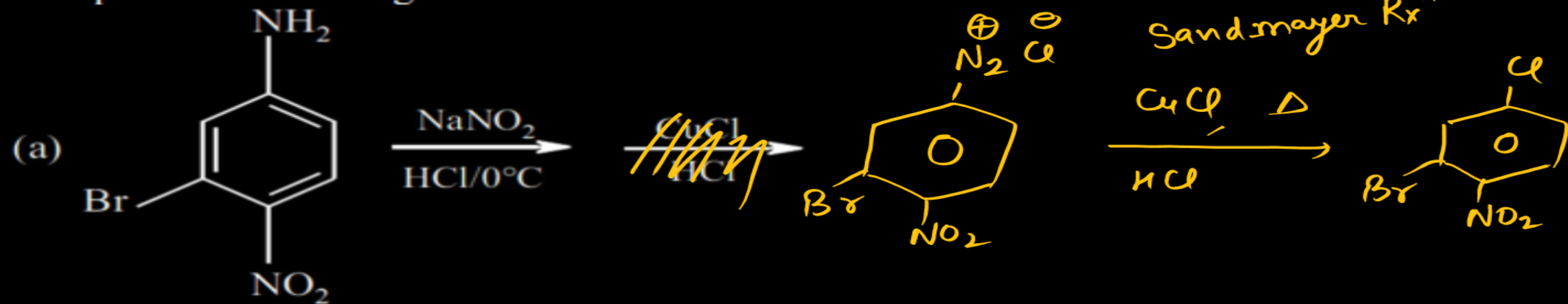
Q. Classify each of the following species as either antiaromatic, aromatic or non aromatic.

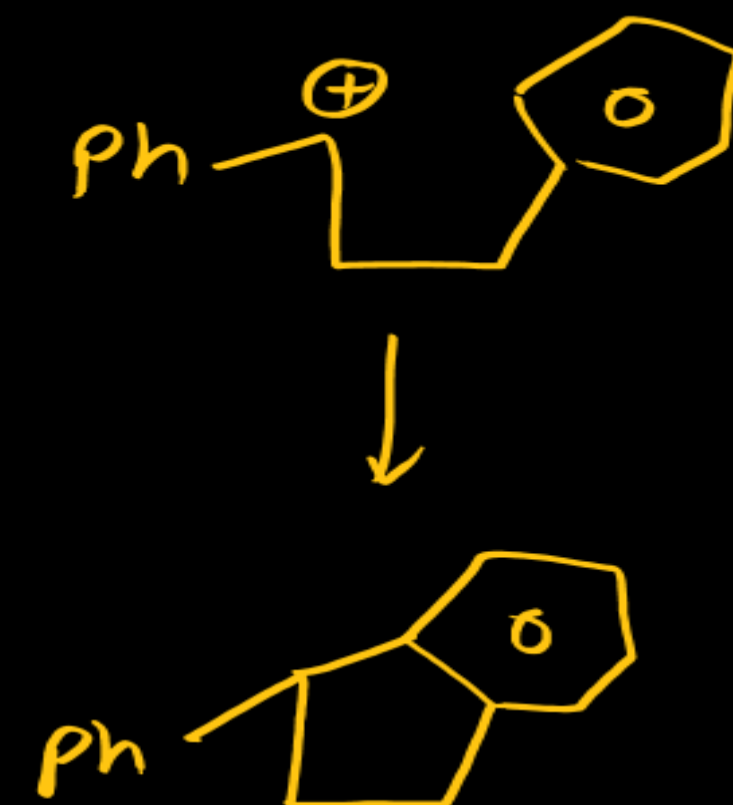
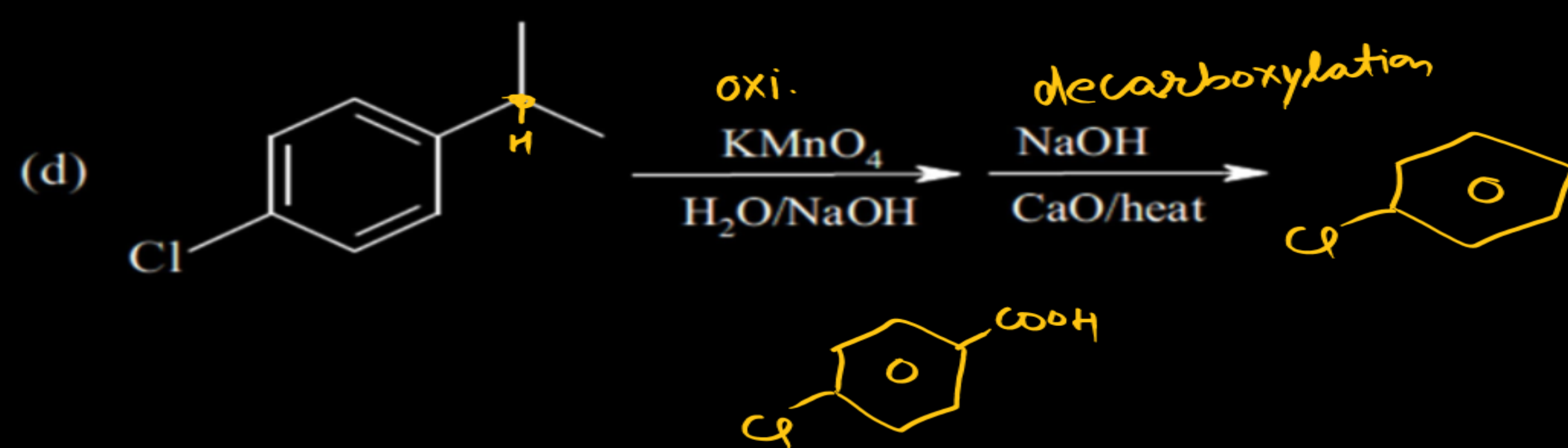


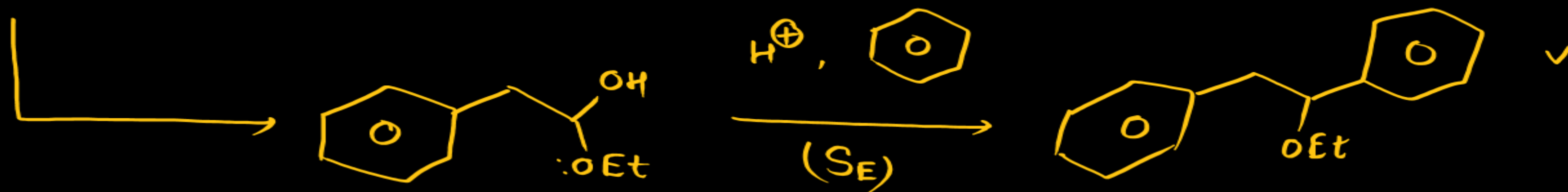
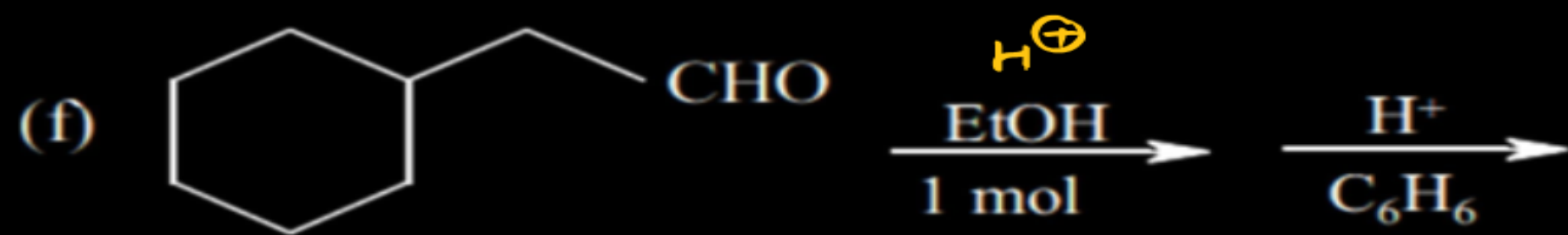
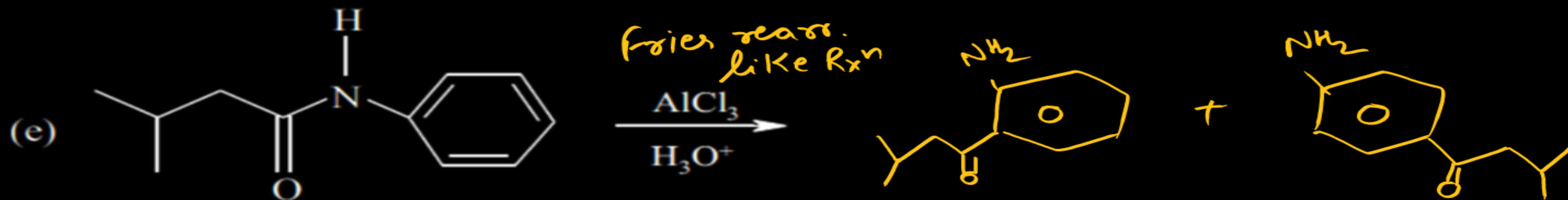
$18\pi e^-$

$2 + 4 + 4 + 4 + \dots$

Q Complete the following reactions:

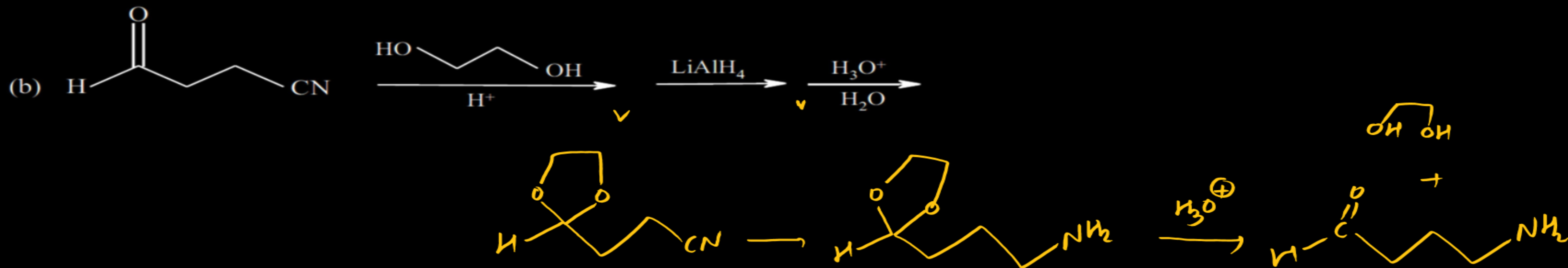
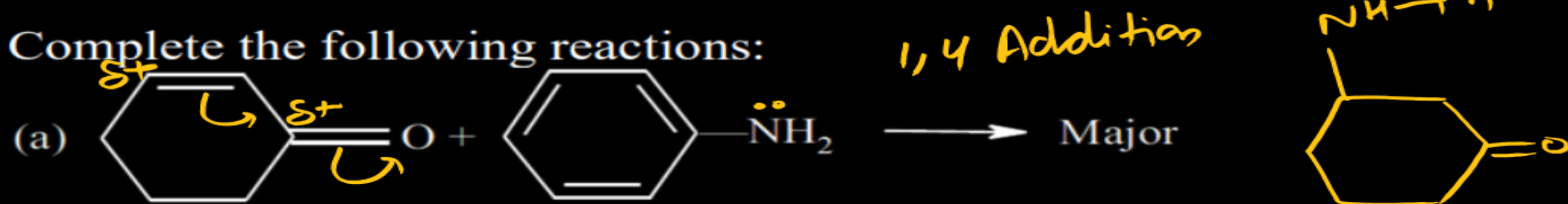


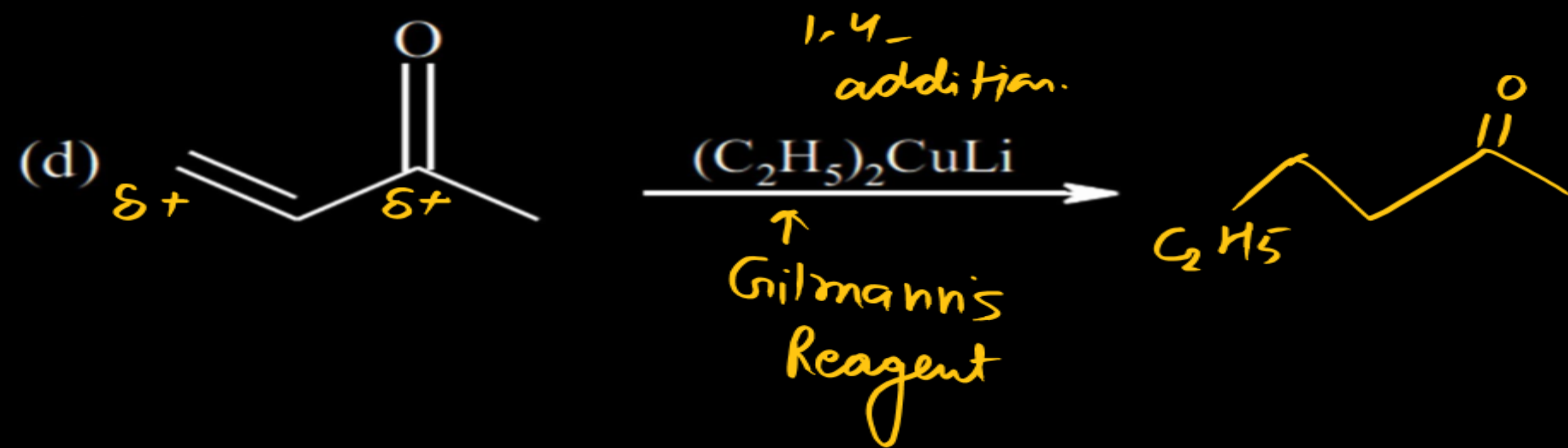
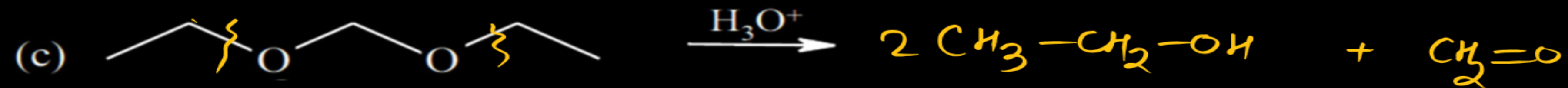


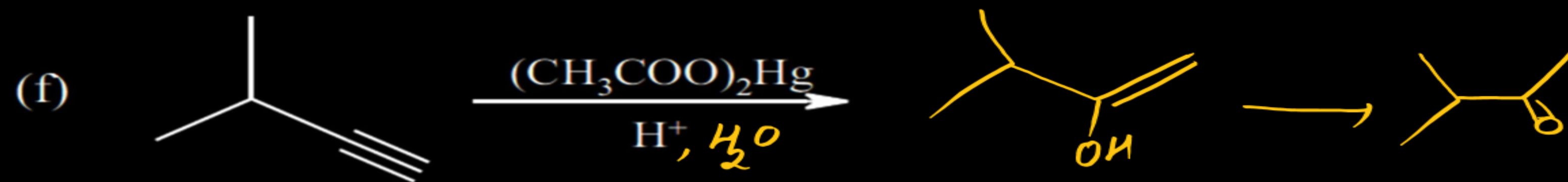
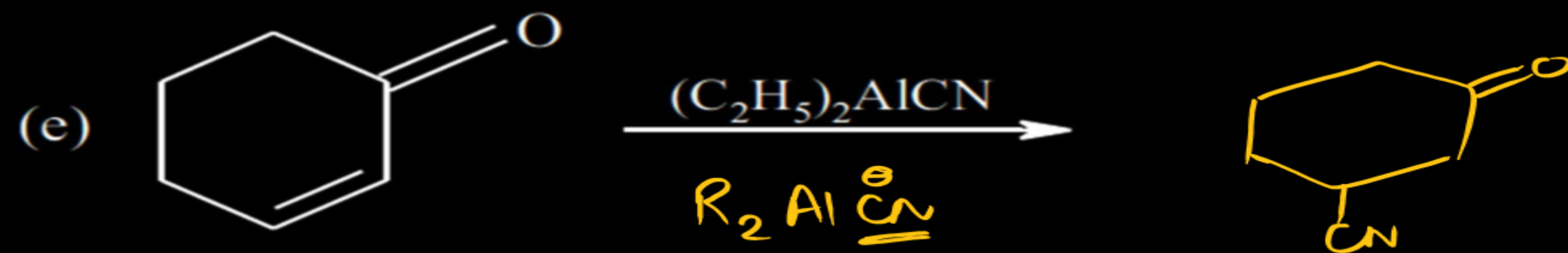


Q

Complete the following reactions:

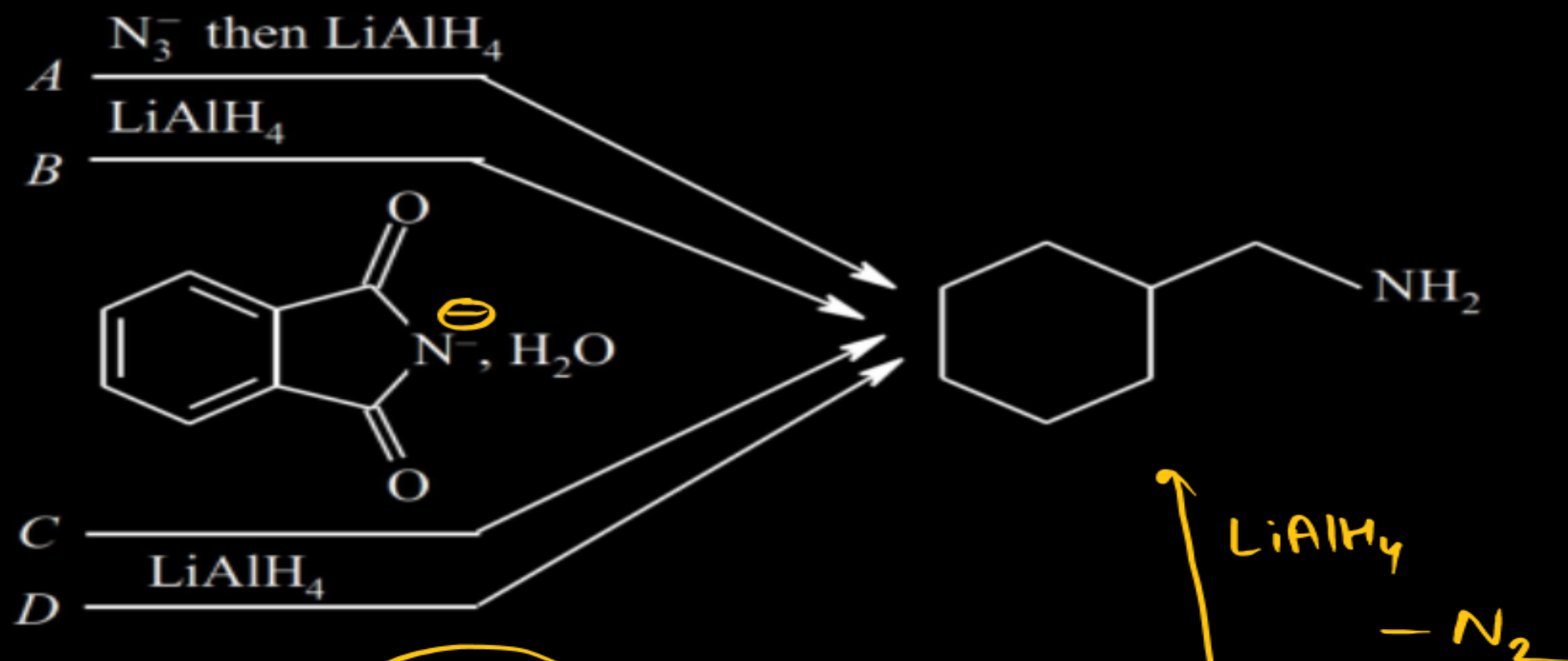




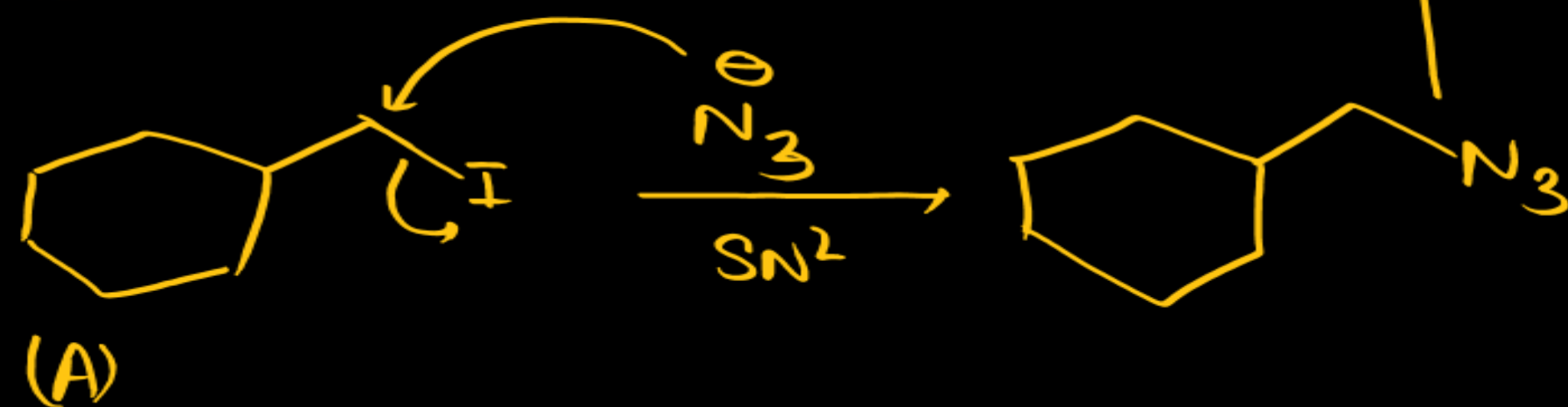


Q

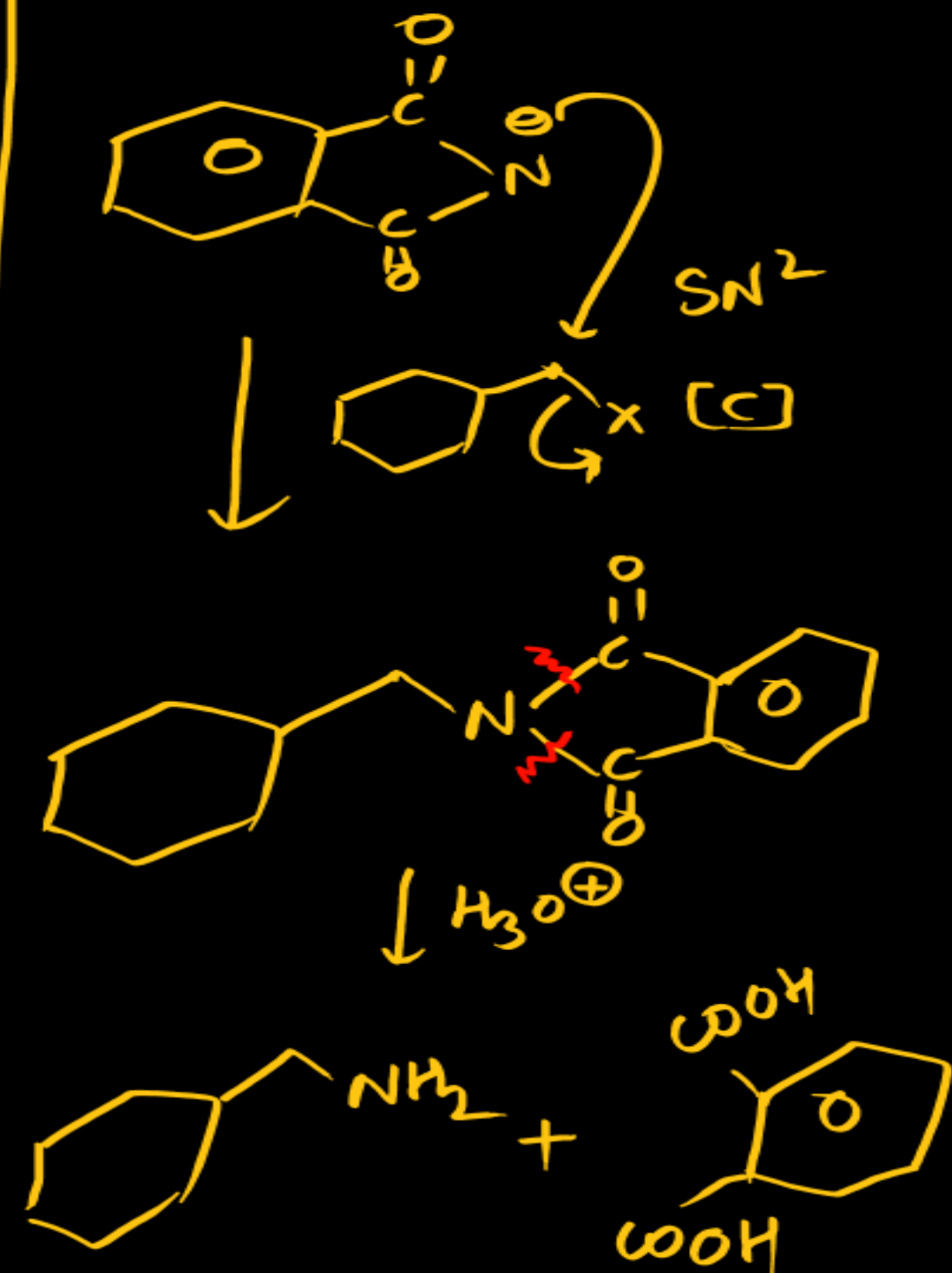
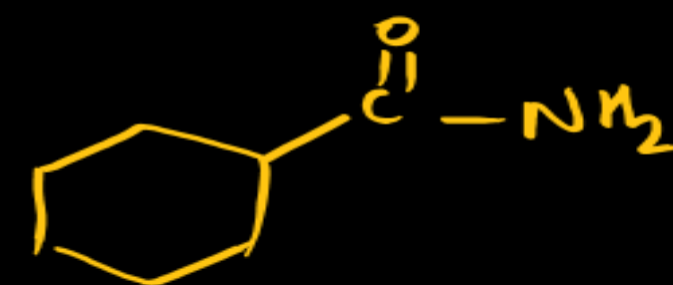
Aminomethylcyclohexane can be prepared from simple starting materials using the four methods shown below. Provide a suitable starting material for each of the reaction shown:



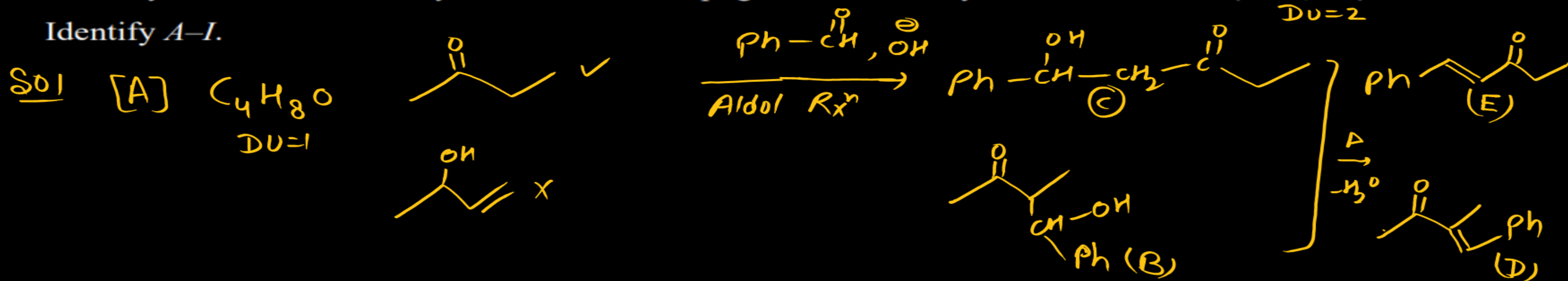
Sol

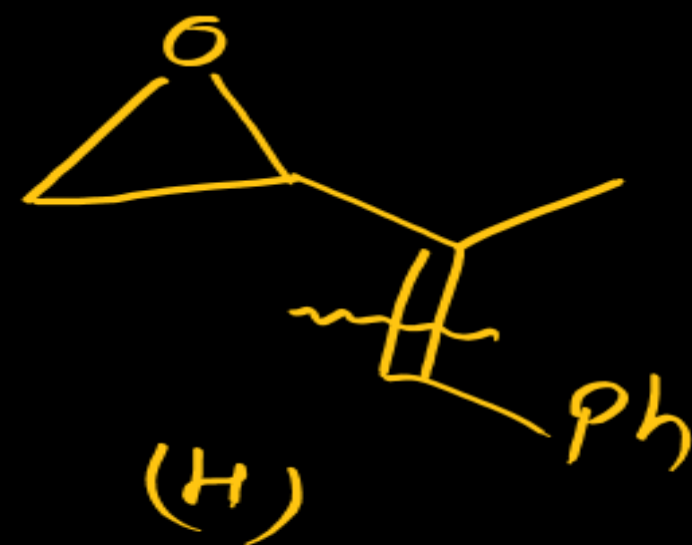
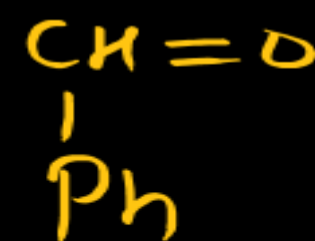
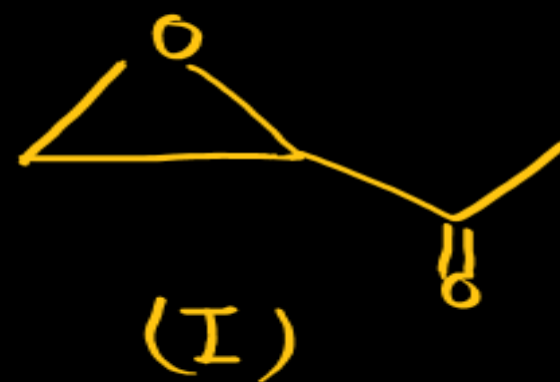
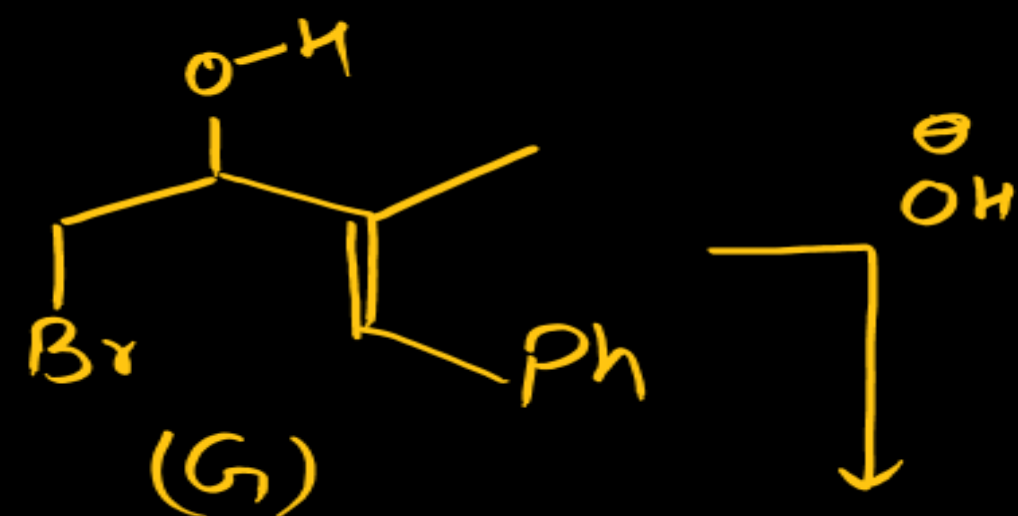
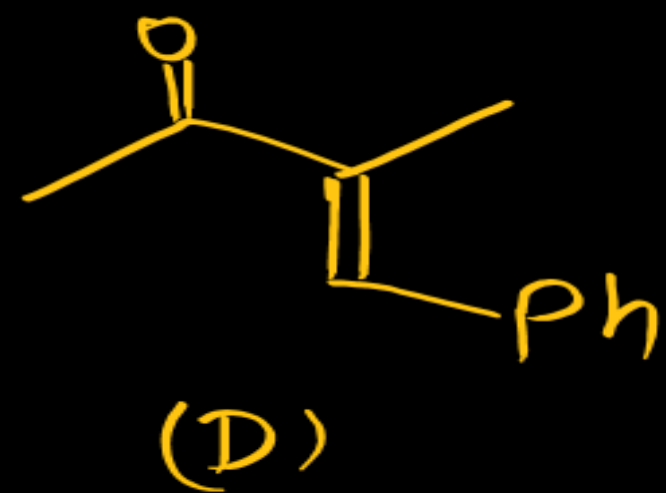


B and D

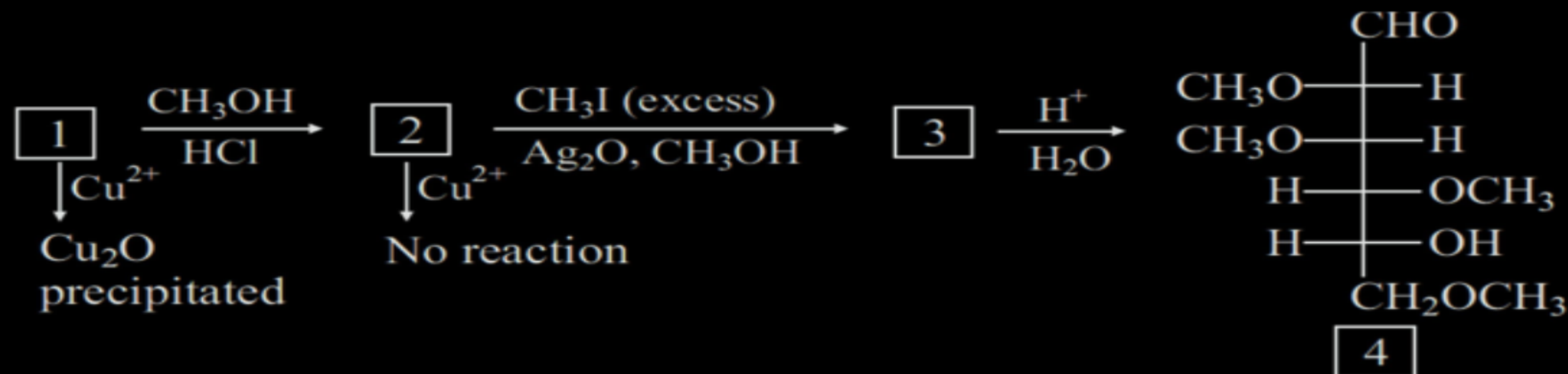


Q Compound $A(C_4H_8O)$ gives positive iodoform test but does not decolourize bromine water solution or it does not evolve any gas on treatment with $NaH^{\oplus}H^{\ominus}$. Reaction of A with excess of benzaldehyde in alkaline medium gives two products B and C . These products are unstable and readily dehydrate into stable products D and E . D reacts with I_2 in basic medium to give iodoform; E does not. Treatment of D with bromine in acetic acid medium yields F . F can be reduced by $NaBH_4$ to give G . Warming G in basic solution results in formation of H . Ozonolysis of H followed by reductive work-up, gives benzaldehyde and a ketone $I(C_4H_6O_2)$.
 Identify $A-I$.





Q. Unknown monosaccharide 1 was converted to tetramethylated aldohexose 4 through the sequence of reaction shown below:



Using the information provided, draw the correct structures of 1, 2 and 3.

